

Qiucheng Wu

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Research Interests

Multimodal LLM: agents & visual understanding; controllable generation in diffusion models

Education

University of California, Santa Barbara, CA 2021 - Present
Ph.D. student in Computer Science, advised by **Prof. Shiyu Chang**

University of Michigan, MI 2017 - 2021
M.S. in Computer Science and Engineering
B.S. in Computer Science and Engineering

Shanghai Jiao Tong University, Shanghai, China 2015 - 2017
B.S. in Electrical and Computer Engineering

Work Experience

Adobe, Research Intern Summer 2023 | Summer 2024 | Summer 2025
Advisor: Dr. Handong Zhao
Research Focus: MLLM Adaptation & Alignment, MLLM Agents, Controllable Vision Generation
Outcomes: *ICCV 2025* (MLLM visual understanding), *ICCV 2023* & *CVPR 2023* (text control in diffusion)

Picsart, Research Intern Summer 2022
Advisor: Dr. Humphrey Shi
Research Focus: Controllable Image Generation
Outcomes: *CPAL 2024*, *TIP 2023*

Fitly, Machine Learning Engineer Intern 2019 - 2020
Advisor: Dr. Honglak Lee
Research Focus: Few-shot Image Classification

Publications and Preprints

‡ indicates co-first author.

- [1] **Qiucheng Wu**, Handong Zhao, Michael Saxon, Trung Bui, William Yang Wang, Yang Zhang, and Shiyu Chang. “VSP: Diagnosing the Dual Challenges of Perception and Reasoning in Spatial Planning Tasks for MLLMs”. *ICCV*, 2025. [\[paper\]](#) [\[code\]](#)
- [2] **Qiucheng Wu**, Handong Zhao, Zhixin Shu, Jing Shi, Yang Zhang, and Shiyu Chang “VividCam: Learning Unconventional Camera Motions from Virtual Synthetic Videos”. *Arxiv*. [\[demo\]](#)
- [3] Xinyi Gao[‡], **Qiucheng Wu**[‡], Yang Zhang[‡], Xuechen Liu, Kaizhi Qian, Ying Xu, and Shiyu Chang. “A Hierarchical Probabilistic Framework for Incremental Knowledge Tracing in Classroom Settings”. *Arxiv*. [\[paper\]](#) [\[code\]](#)
- [4] **Qiucheng Wu**[‡], Yifan Jiang[‡], Junru Wu[‡], Kai Wang, Gong Zhang, Humphrey Shi, Zhangyang Wang, and Shiyu Chang. “Decoding Micromotion in Low-dimensional Latent Spaces from StyleGAN”. *CPAL*, 2024. [\[paper\]](#) [\[code\]](#) [\[demo\]](#)
- [5] **Qiucheng Wu**[‡], Yujian Liu[‡], Handong Zhao, Trung Bui, Zhe Lin, Yang Zhang, and Shiyu Chang. “Harnessing the spatial-temporal attention of diffusion models for high-fidelity text-to-image synthesis”. *ICCV*, 2023. [\[paper\]](#) [\[code\]](#)
- [6] **Qiucheng Wu**, Yujian Liu, Handong Zhao, Ajinkya Kale, Trung Bui, Tong Yu, Zhe Lin, Yang Zhang, and Shiyu Chang. “Uncovering the disentanglement capability in text-to-image diffusion models”. *CVPR*, 2023. [\[paper\]](#) [\[code\]](#) [\[demo\]](#)

- [7] **Qiucheng Wu**[‡], Yifan Jiang[‡], Junru Wu[‡], Victor Kulikov, Vidit Goel, Nikita Orlov, Humphrey Shi, Zhangyang Wang, and Shiyu Chang. “Broad Spectrum Image Deblurring via An Adaptive Super-Network”. **IEEE TIP**, 2023. [\[paper\]](#) [\[code\]](#)
- [8] Yimeng Zhang[‡], Akshay Karkal Kamath[‡], **Qiucheng Wu**[‡], Zhiwen Fan[‡], Wuyang Chen, Zhangyang Wang, Shiyu Chang, Sijia Liu, and Cong Hao. “Data-Model-Circuit Tri-Design for Ultra-Light Video Intelligence on Edge Devices”. **ASP-DAC**, 2023. [\[paper\]](#)
- [9] Janak Sharda, Wantong Li, **Qiucheng Wu**, Shiyu Chang, and Shimeng Yu. “Temporal frame filtering for autonomous driving using 3D-stacked global shutter CIS with IWO buffer memory and near-pixel compute”. **IEEE TCAS-I**, 2023. [\[paper\]](#)
- [10] Wantong Li, **Qiucheng Wu**, Janak Sharda, Shiyu Chang, and Shimeng Yu. “Temporal Frame Filtering with Near-Pixel Compute for Autonomous Driving”. **AICAS**, 2022. **ASP-DAC**, 2023. [\[paper\]](#)
- [11] Nina Zhou, **Qiucheng Wu**, Zewen Wu, Simeone Marino, and Ivo D Dinov. “DataSifterText: Partially synthetic text generation for sensitive clinical notes”. **Journal of Medical Systems**, 2022. [\[paper\]](#)
- [12] Yijie Guo, **Qiucheng Wu**, and Honglak Lee. “Learning action translator for meta reinforcement learning on sparse-reward tasks”. **AAAI**, 2022. [\[paper\]](#)
- [13] Simeone Marino, Nina Zhou, Yi Zhao, Lu Wang, **Qiucheng Wu**, and Ivo D Dinov. “DataSifter: statistical obfuscation of electronic health records and other sensitive datasets”. **Journal of Statistical Computation and Simulation**, 2019. [\[paper\]](#)

Service

- **Local organization chair**, Southern California NLP Workshop (SoCalNLP 2022)
- **Reviewer**, NeurIPS, ICML, ICCV, CVPR, TMLR

Teaching Experience

- **Teaching Assistant** at University of California, Santa Barbara Fall 2025, 2024, 2023, 2021
Intermediate Python
Machine Learning
- **Teaching Assistant** at University of Michigan Fall 2021, 2020
Introduction to Artificial Intelligence
- **Teaching Assistant** at Shanghai Jiao Tong University Fall 2016
Applied Honors Calculus

Selected Awards

- James B. Angell Scholar, University of Michigan 2019
- Capstone Gold Prize, Shanghai Jiao Tong University 2019
- MIDAS Data Science Symposium, Most Interesting Methodological Advances, Univsity of Michigan 2018
- MIDAS Data Science Symposium, Most Likely Health Impact, Univsity of Michigan 2018
- Bosch Scholarship, Shanghai Jiao Tong University 2016
- Academic Excellence Scholarship, Shanghai Jiao Tong University 2016